SOC542 Statistical Methods in Sociology II

Multiple Regression

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Plan

- Recap
- Multiple regression: An overview
- Lab: Multiple regression in R

Recap

What we have learned so far

1. Fundamentals of frequentist inference
2. Simple linear regression
3. Probability and Bayesian inference

Multiple regression

OLS assumptions review

- x and y are independently and identically distributed (IID).
  - The sample x must contain some variability. Specifically, $var(x) > 0$.
- The conditional distribution of u given x has a mean of zero.
  - Errors are independent $E[u_i|x_i] = E[u_i] = 0$.
  - Errors have constant variance $var(u_i) = \sigma^2$.
  - Errors are uncorrelated.
- If these assumptions are met, then OLS is **BLUE**
  - The **Best Linear conditionally Unbiased Estimator**

Multiple regression

Simple linear regression

- Let's say we estimate a simple linear regression of the form:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{u}$$

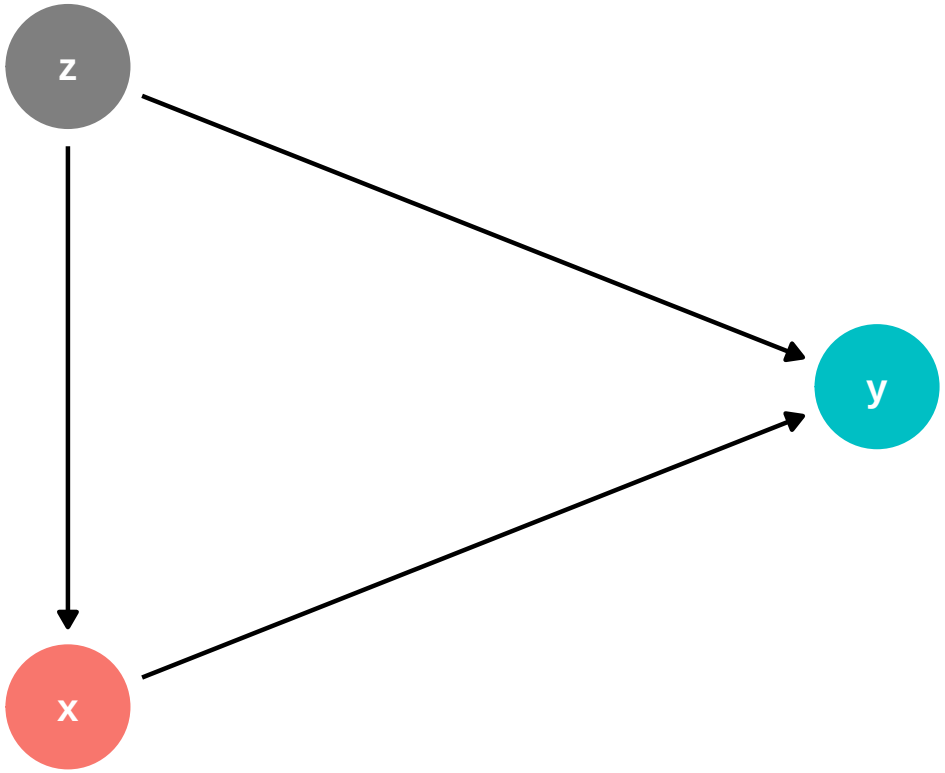
- In this case, we assume that the outcome y is a linear function of a single predictor x.
- But what if we think have reason to believe that y is also a function of other predictors?

Multiple regression

Omitted variable bias

- Omitted variable bias occurs when we leave out, or *omit*, a predictor that should be in our model.
- It exists when
  - x is correlated with the omitted variable z.
  - The omitted variable is a predictor of the dependent variable y.

Omitted variable bias



Multiple regression

Consequences omitted variable bias

- The assumption that $E(u_i|x_i) = 0$ is violated.
  - If z is correlated with x but not included, then the error term u captures the unmeasured effect of z *and* thus u is correlated with x.
- The slope coefficient β_1 will be *biased*.
  - The mean of the sampling distribution of the OLS estimator may not equal the true effect of x.
  - $\hat{\beta}_1 = \beta_1 + bias$
- The OLS estimator is *inconsistent* as $\hat{\beta}_1$ does not converge in probability to β_1.
  - Bias remains even with large samples.
- The greater the correlation between x and u, the greater the bias.

Multiple regression

Example: Weath and education

$$Wealth_i = \beta_0 + \beta_1 Educ + u$$

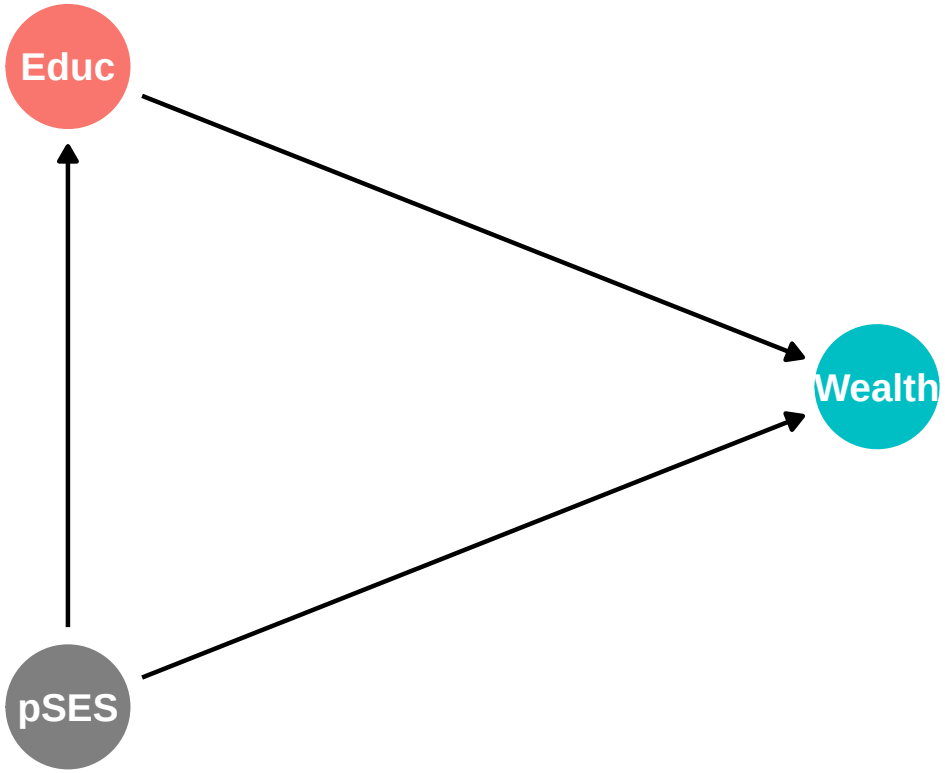
- Assume a strong, positive relationship between education and wealth (i.e. $\hat{\beta}_1$ is positive)
  - What other factors might be correlated with education *and* predict wealth?

Multiple regression

Example: Weath and education

- Parental socioeconomic status (SES) is correlated with education and predicts wealth.
  - The estimate of the effect of education is thus *biased* without taking parental SES into account.

Drawing the DAG



Simulating omitted variable bias

```
N <- 100
x <- rnorm(N,2,1)
z <- 0.2*x + rnorm(N,3,1)
y <- 0.8*x + -2*z + rnorm(N, 0, 1)
```

Simulating omitted variable bias

```
m.omit <- lm(y ~ x)
m.both <- lm(y ~ x + z)
m.omit$coefficients %>% round(2) %>% print()
```

```
## (Intercept)           x
##       -5.89        0.48
```

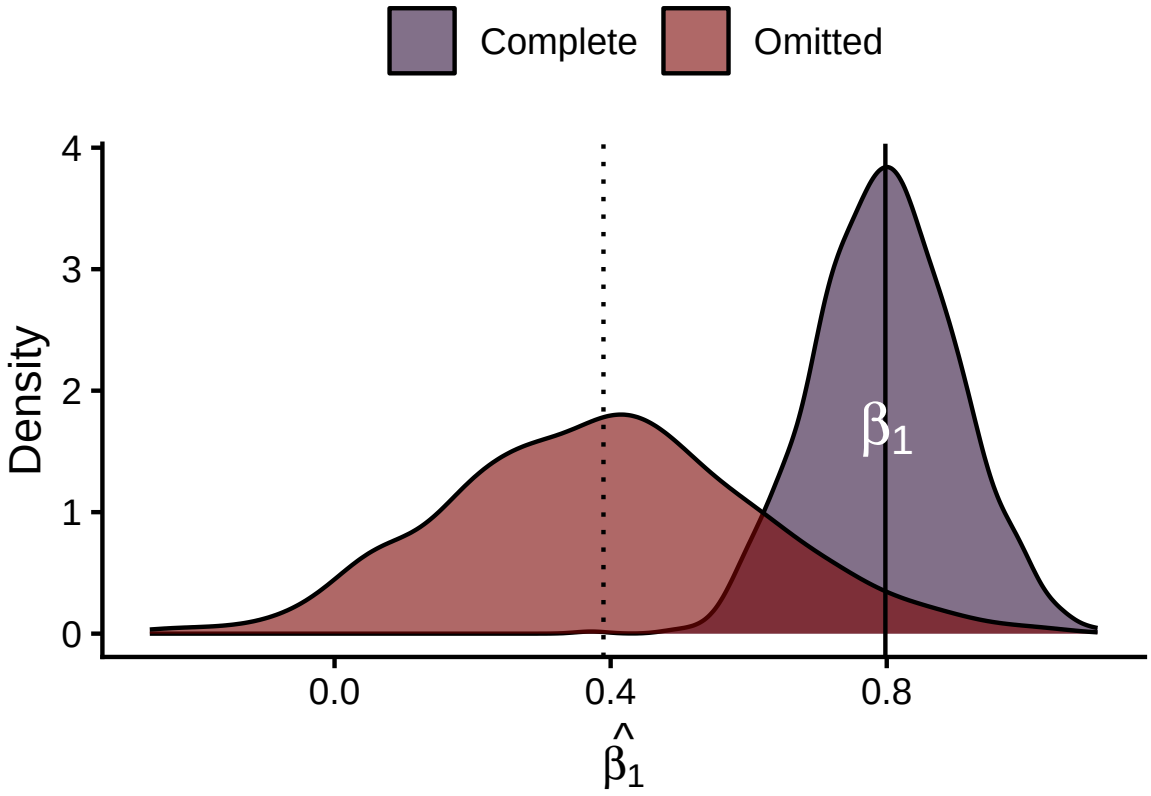
```
m.both$coefficients %>% round(2) %>% print()
```

```
## (Intercept)           x           z
##        0.22        0.83       -2.06
```

Simulating omitted variable bias

```
coefs.omitted <- c()
coefs.complete <- c()
sims <- 1E3
for (i in 1:sims) {
    x <- rnorm(N,2,1)
    z <- 0.2*x + rnorm(N,3,1)
    y <- 0.8*x + -2*z + rnorm(N, 0, 1)
    m.omit <- lm(y ~ x)
    m.both <- lm(y ~ x + z)
    coefs.omitted[i] <- m.omit$coefficients[2]
    coefs.complete[i] <- m.both$coefficients[2]
}
```

Simulating omitted variable bias



Distribution of $\hat{\beta}_1$ over 1000 simulations.

Dotted line shows estimate when z is omitted; solid line when z is included in the regression.

Multiple regression

The multiple regression model

- In a multiple regression model, we specify a linear relationship between an outcome and a set of k predictors.

$$E[y|x_1, x_2, ..., x_k] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + u$$

Multiple regression

Independent variables and controls

- The predictors added to the model can be considered as additional **independent variables** or as **controls**.
- In general, we use the former term when we have a theoretical reason to be interested in a effect of a variable and the latter when we expect it to matter but are not interested in analyzing the relationship directly.
- We typically add control variables to mitigate omitted variable bias.

Multiple regression

Interpreting coefficients

- Consider the following population model:

$$y_i = \beta_0 + \beta_1 x + \beta_2 z + u$$

- β_1 is the effect of a unit change in x *when* z *is held constant*.

$$\beta_1 = \frac{\Delta y}{\Delta x}, \textit{holding } z \textit{ constant}$$

Multiple regression

Interpreting the intercept

- Consider the same model:

$$y_i = \beta_0 + \beta_1 x + \beta_2 z + u$$

- β_0 is the expected value of y_i when $x = 0$ and $z = 0$.

Multiple regression

The OLS estimator

- The OLS estimator minimizes the sum of the squared residuals.
- Over n observations and k predictors, we minimize the following quantity:

$$\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_{1i} - \beta_2 x_{2i} - ... - \beta_k x_{ki})^2$$

- Thus, the predicted values are

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + ... + \hat{\beta}_k x_{ki}$$

- And the residuals are defined as $\hat{u}_i = y_i - \hat{y}_i$ for $i = 1, ..., n$.

Multiple regression

OLS in matrix form

- We can write the equation for multiple regression more compactly using matrix notation. Here X is an n by k matrix of predictors and $\boldsymbol{\beta}$ is vector of coefficients with length k.

$$y = \beta_0 + \boldsymbol{\beta} X + u$$

- Ordinary least squares estimates can be computed directly using matrix multiplication, where X is a matrix of predictors and the first column is a vector of 1s (for the intercept) and y is the outcome.

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T y$$

Multiple regression

OLS in matrix form

```
Intercept <- rep(1,N)
X <- cbind(Intercept,x,z)

Betas <- solve(t(X) %*% X) %*% (t(X)  %*% y)
print(t(Betas))
```

```
##       Intercept         x         z
## [1,] -0.6282864 0.8244212 -1.825717
```

```
m <- lm(y ~ x + z)
print(m$coefficients)
```

```
## (Intercept)           x           z
##  -0.6282864   0.8244212  -1.8257171
```

`t()` is the transpose operation, `solve()` finds the inverse of a matrix, and `%*%` is the matrix multiplication operator.

Multiple regression

OLS in matrix form

- OLS estimates are derived directly from algebraic manipulation of the data.
- OLS is a special case. Other approaches we will encounter in a few weeks require more complicated *maximum likelihood estimation*.
- Bayesian regression with uniform priors will converge to the least squares solution, despite a radically different estimation procedure.

Multiple regression

Model fit and the Standard Error of the Regression

- The **Standard Error of the Regression (SER)** is an estimate of the standard deviation of the error term u_i. It captures the spread of y around the regression line.
- For a single regressor,

$$SER = \sqrt{\sigma_{\hat{u}}} = \sqrt{\frac{1}{n-2}\sum_{i=1}^{n}\hat{u}_i^{\,2}} = \sqrt{\frac{SSR}{n-2}}$$

- $n-2$ accounts for degrees of freedom used by slope and intercept.
- A smaller SER indicates better fit.

Multiple regression

Model fit and the Standard Error of the Regression

- If we have multiple predictors we need to include an degrees of freedom adjustment k, where k is the number of predictors. The -1 accounts for the intercept.

$$SER = \sqrt{\sigma_{\hat{u}}} = \sqrt{\frac{1}{n-k-1}\sum_{i=1}^{n}\hat{u}_i^2} = \sqrt{\frac{SSR}{n-k-1}}$$

- The adjustment has a small effect when n is large.

Multiple regression

Model fit and R^2

- We define R^2 in the same way as a simple regression:

$$R^2 = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = \frac{ESS}{TSS}$$

$$R^2 = 1 - \frac{SSR}{TSS}$$

Multiple regression

Adjusted R^2

- R^2 increases mechanistically as we add predictors because the SSR declines as long as $\hat{\beta}_k \neq 0$, inflating model fit.
- A degrees of freedom correction is used to adjust for this:

$$Adjusted\ R^2 = 1 - \frac{n-1}{n-k-1}\frac{SSR}{TSS}$$

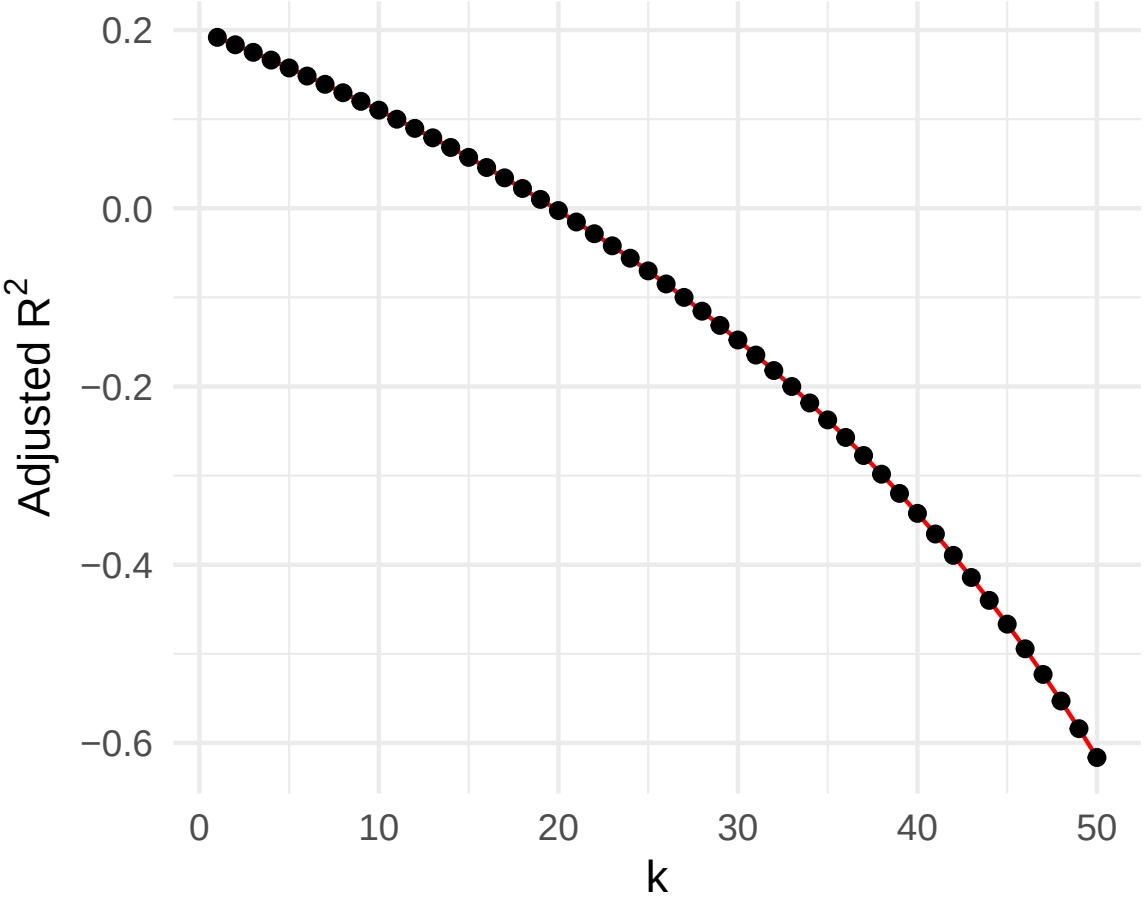
Multiple regression

Properties of adjusted R^2

- Adjusted R^2 is *always less than* R^2.
- Adding a predictor can increase Adjusted R^2, but it can decline if the change to the SSR is weaker than the offset $(n-1)/(n-k-1)$.
- Adjusted R^2 can be negative if the reduction in SSR does not offset $(n-1)/(n-k-1)$.

Multiple regression

The penalty $\frac{n-1}{n-k-1}$ increases as we add predictors



This example shows the effect of the degree of freedom adjustment, assuming $\beta_k = 0$ for all $k > 1$.

Multiple regression

Bayesian R^2

- There is no direct analogue for R^2 in Bayesian statistics
  - Recall that frequentist models assume *fixed* parameters, whereas Bayesian parameters have *distributions*.
- If we treat the Bayesian estimates as fixed, for example by taking the median of the posterior distribution $\hat{\beta}_k$, we could calculate something using the formula above, but it would not account for the *uncertainty* contained in the posterior distribution.

Multiple regression

Bayesian R^2

- Instead, we use posterior simulations to repeat the calculation across all samples from the posterior.
- Bayesian R^2 therefore has a posterior distribution.[1] We can summarize this into a single metric using the same approach as the regression coefficients, e.g. using the median of the posterior distribution.[2]

[1] "Everything that depends upon parameters has a posterior distribution" - McElreath 98.

[2] See GHV 170-171

Multiple regression

Significance tests: t-tests

- Like simple linear regression, we can interpret the statistical significance of regression coefficients using the t-statistics.
- Typically, we are interested in testing the null hypothesis that $\beta_k = 0$. We get the t-statistic by dividing a coefficient by its standard error:

$$t = \frac{\hat{\beta}_k - 0}{SE(\hat{\beta}_k)} = \frac{\hat{\beta}_k}{SE(\hat{\beta}_k)}$$

- We can the use the t-statistic to look up the relevant *p-value*.

Multiple regression

Confidence interval

- Most regression software provides a 95% confidence interval around each estimate. For β_j this would take the following form:

$$[\hat{\beta}_j - 1.96SE(\hat{\beta}_j), \hat{\beta}_j + 1.96SE(\hat{\beta}_j)]$$

- Recall that only 5% of the probability density of a t-distribution is greater than $|1.96|$.

Multiple regression

Joint tests

- The F-statistic is used to test a **joint hypothesis**.
- If we consider a two variable example, we might test the following *null hypothesis*:

$$H_N : \beta_1 = 0, \beta_2 = 0$$

- A joint test has q restrictions. In this case, $q = 2$.
- The *alternative hypothesis* H_A is that one or more of the q restrictions does not hold.

Multiple regression

Joint tests and the F-statistic

- Since we expect the predictors to have a *joint sampling distribution*, we cannot conduct a joint test by summarizing a series of paired tests (e.g. a t-test for every predictor) because the t-statistics are not independent.
- Instead, we must calculate the F-statistic. Where $q = 2$ it is defined as:

$$F = \frac{1}{2}\left(\frac{t_1^2 + t_2^2 - 2\hat{\rho}_{t_1,t_2} t_1 t_2}{1 - \hat{\rho}_{t_1,t_2}^2}\right)$$

Multiple regression

Joint tests and the F-statistic

- If $\hat{\rho}_{t_1,t_2} = 0$ the equation simplifies to the average of the squared t-statistics:

$$F = \frac{1}{2}(t_1^2 + t_2^2)$$

- The p-value can then be derived from the relevant *F-distribution*, where $F \sim F_{q,\infty}$
- Typically, we use an F-test to test the restriction that $\beta_1 = 0, \beta_2 = 0, ..., \beta_k = 0$.

Multiple regression

Joint tests and the F-statistic

- If we assume the residuals are *homoskedastic*, we can test the restriction $\beta_1 = 0, \beta_2 = 0, ..., \beta_k = 0$ using the following formula:

$$F_0 = \frac{(SSR_r - SSR_u)/q}{SSR_u/(n-k-1)}$$

- The SSR_r is obtained from *restricted* model where we calculate the SSR assuming the null hypothesis is true. The SSR from the fitted model, SSR_u, is known as the *unrestricted* SSR.
- The test statistic is assessed using an *F-distribution* with q degrees of freedom and $n - k - 1$ observations.
- In most cases the homoskedasticity assumption is likely violated, so we use the more complicated formula from the previous slide, known as the *heteroskedasticity robust* F-statistic.

Multiple regression

Interpreting regression output

```
Call:
lm(formula = y ~ x + z)

Residuals:
     Min       1Q   Median       3Q      Max
-2.26702 -0.71764  0.08484  0.67756  2.66094

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.62829    0.37897  -1.658    0.101
x            0.82442    0.10177   8.101 1.64e-12 ***
z           -1.82572    0.09985 -18.285  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.031 on 97 degrees of freedom
Multiple R-squared:  0.7897,    Adjusted R-squared:  0.7853
F-statistic: 182.1 on 2 and 97 DF,  p-value: < 2.2e-16
```

Multiple regression

Bayesian approaches

- "We have essentially no interest in using hypothesis tests for regression because we almost never encounter problems where it would make sense to think of the coefficients as being exactly zero" - GHV 147
- Bayesian regression is assessed by analyzing the posterior distribution of parameters to understand uncertainty.
- Nonetheless, Bayesian equivalents to t-tests and F-tests can be used if desired.[3]

[3] See Kruschke and Liddell 2018.

Multiple regression

Multicollinearity

- **Multicollinearity** occurs when a predictor x is highly correlated one or more other predictors z.
  - **Perfect multicollinearity** arises when $cor(x, z) = 1$ or -1.
    - Usually due to some type of misspecification. e.g. accidentally including the same variable twice.
  - **Imperfect multicollinearity** means that two or more regressors are highly correlated.

Multiple regression

Multicollinearity and its implications

- Assume the following model and that x and z are highly correlated:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \hat{\beta}_2 z_i + u_i$$

- The variance of $\hat{\beta}_1$[4] is inversely proportional to $1 - \rho^2_{x,z}$, where $\rho_{x,z}$ is the correlation between x and z.
  - If $\rho_{x,z}$ is large, then this term is small and thus the variance is large.
- Multicollinearity *increases variance* and *reduces precision*, potentially making β_1 **non-identifiable**.

[4] The same issue also applies to $\hat{\beta}_2$

Simulating multicollinearity

```
N <- 100
x <- rnorm(N,2,1)
x2 <- rnorm(N,0,1)
z <- 0.5*x + rnorm(N,0,1)
y <- 0.5*x + -0.5*x2 + 0.05*z + rnorm(N, 0, 2)
```

Simulating multicollinearity

```r
m1 <- summary(lm(y ~ x + x2))
m2 <- summary(lm(y ~ x + x2 + z))
round(m1$coefficients,2) # omitted variable bias
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.01       0.45   -0.03     0.98
## x               0.50       0.20    2.47     0.02
## x2             -0.58       0.19   -3.02     0.00
```

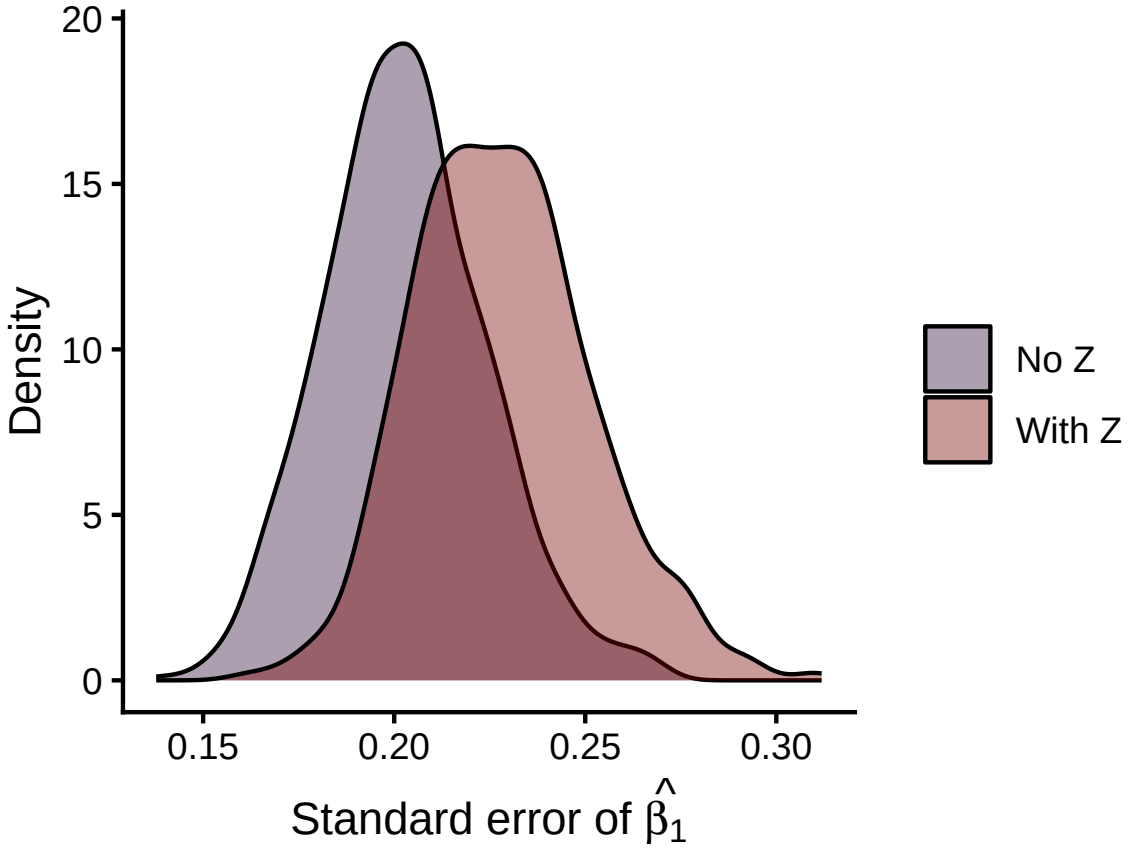
```r
round(m2$coefficients,2) # multicollinearity
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.01       0.46   -0.02     0.99
## x               0.43       0.22    1.91     0.06
## x2             -0.56       0.19   -2.92     0.00
## z               0.13       0.18    0.73     0.47
```

Simulating multicollinearity

```
se.omitted <- c()
se.complete <- c()
sims <- 1E3
for (i in 1:sims) {
    x <- rnorm(N,2,1)
    x2 <- rnorm(N,0,1)
    z <- 0.5*x + rnorm(N,0,1)
    y <- 0.5*x + -0.5*x2 + 0.05*z + rnorm(N, 0, 2)
    m.omit <- summary(lm(y ~ x + x2))
    m.complete <- summary(lm(y ~ x + x2 + z))
    se.omitted[i] <- m.omit$coefficients[2,2]
    se.complete[i] <- m.complete$coefficients[2,2]
}
```

Simulating multicollinearity



Distribution of standard error over 1000 simulations.

Multiple regression

Addressing multicollinearity

- In general, multicollinearity is less severe than omitted variable bias.
  - The inflated variance will lead to more Type II errors than Type I errors.
  - Omitted variable bias can produce Type I errors, sign errors, and magnitude errors.

Multiple regression

Fixing multicollinearity

- *Solution 1*: Use more data. If we have a larger sample then we might be able to learn from additional variation in x and z.
- *Solution 2*: If we are only concerned about x then we could exclude z. But this risks omitted variable bias if z is also a predictor of y.
- *Solution 3*: Transform or combine predictors (e.g. factor analysis).

Multiple regression

Revisiting our assumptions

- $E(u_i|x_{1i}, x_{2i}, ..., x_{ki}) = 0$
- All $y_i, x_{1i}, x_{2i}, .., x_{ki}$ are IID.
- Large outliers are unlikely.
- No perfect multicollinearity.

Variable selection[5]

- It is conventional practice to include a wide array of potential confounders in a regression model as controls.
  - Sometimes derided as "kitchen sink" or "garbage can" regressions).
- But omitted variable bias, multicollinearity, and collider bias must be considered when deciding what to include.
- Domain knowledge and theory can guide model specifications, but these issues cannot be identified from the data alone.

[5] See Cinelli et al. 2022; Kohler et al. 2020; Keele et al. 2023.

Variable selection

- "[A]uthors should not just list covariates that may affect the outcome but should disclose how they think those covariates and the exposure are causally related."

  Kohler et al. 2020, p. 12
- DAGs are a useful tool for representing our assumptions, guiding variable selection, and identifying problematic specifications.

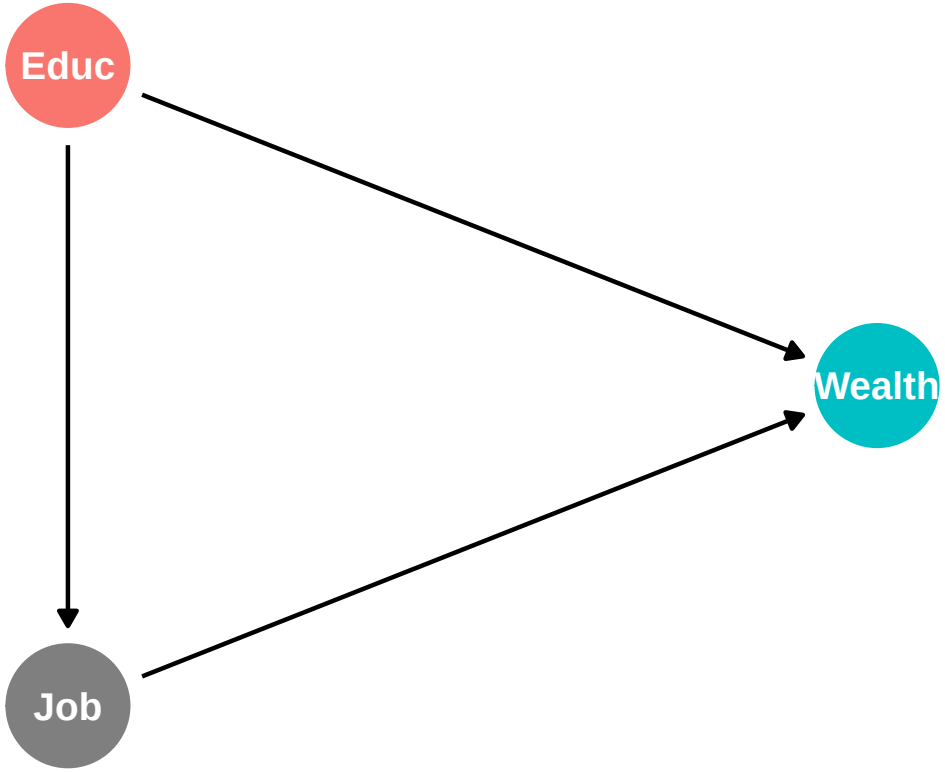
Variable selection

Mediators

- *Mediators* are variables that partially explain the causal effect of x on y
  - e.g. **first job** might mediate explain some of the effect of education on wealth.
- Controlling for a mediator will change the estimate of x from the *total* effect to the *conditional effect*

Variable selection

Mediators



Variable selection

Spurious relationships and confounding

- Sometimes we observe **spurious** relationships in regression models where a correlation between two variables exists, despite the absence of any causal relationship.
  - e.g. Finding that hurricanes with female names *caused* more deaths than male named hurricanes.
- Sometimes this occur purely due to chance, but it can be due to **confounding**: a confounding variable z influences both x and y.

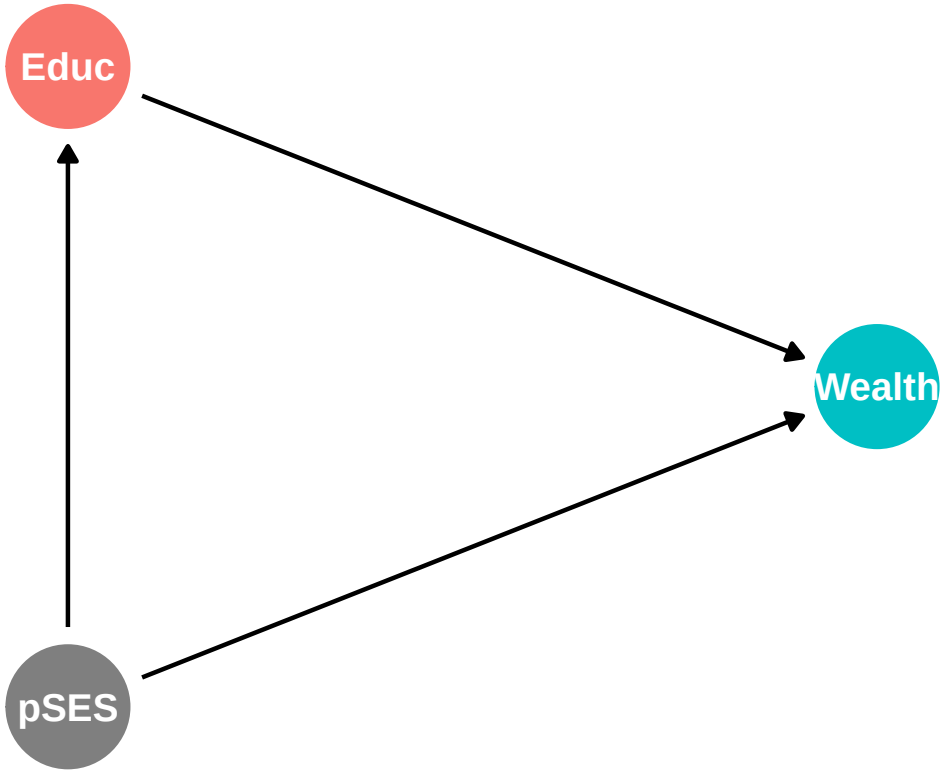
Variable selection

Spurious relationships and confounding

- Adding more predictors can often help to reduce the risk of spurious associations.
  - If we control for the confounder z, the spurious relationship between y and x disappears.
- This is sometimes referred to as "blocking the backdoor path"
- This is related to OVB, which addresses bias in the estimation, rather than the causal structure.

Confounding

Blocking the backdoor path



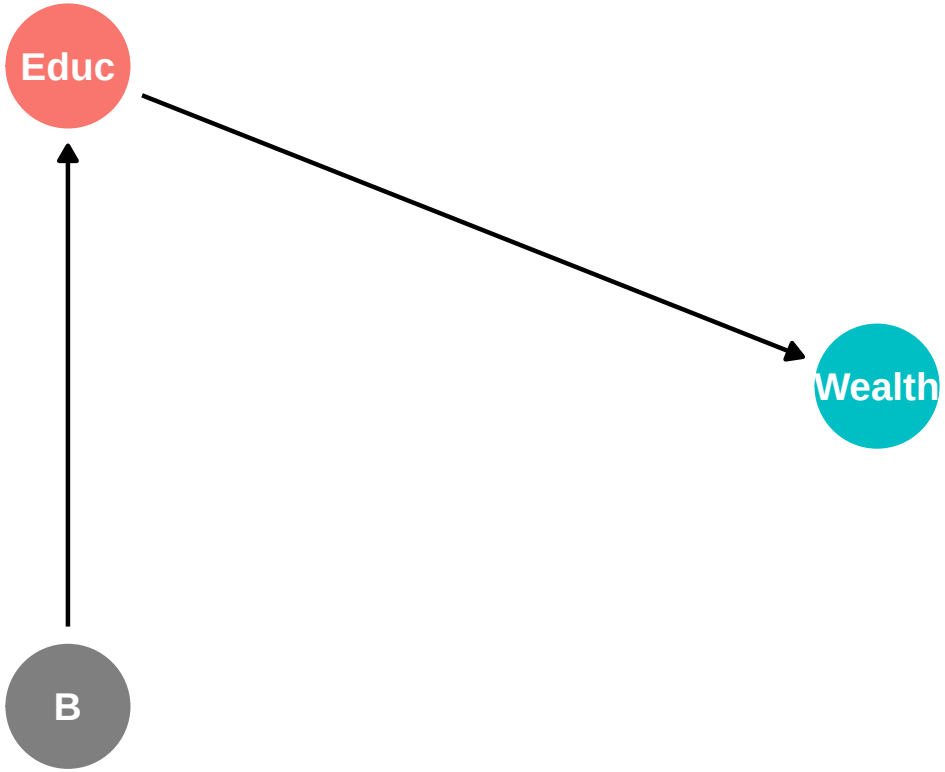
Variable selection

Masked relationships

- Assume a true relationship between y and x.
- We estimate a model $y = \beta_0 + \beta_1 x$.
- The results do not show evidence of an association (i.e. $\hat{\beta}_1 \approx 0$).
- We estimate a second model including a new predictor z.
- Controlling for z allows us to observe a relationship between y and x.

Bad control

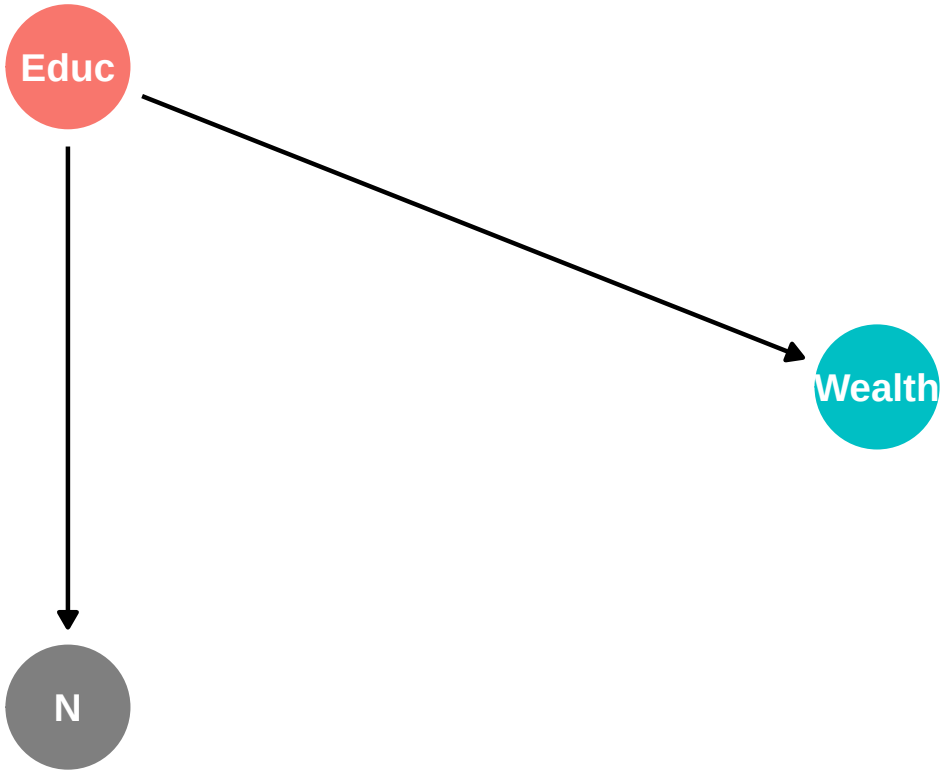
Causes of the independent variable can reduce precision[6]



[6] But we will see later in the semester how these can be helpful for causal indentification as instrumental variables.

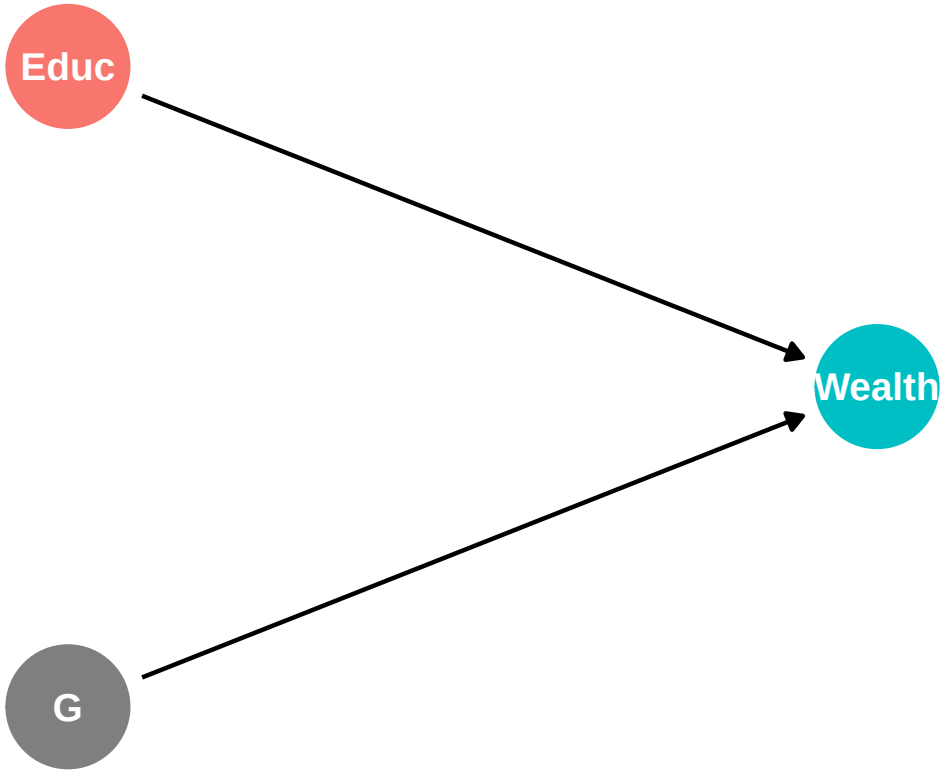
Neutral control

Descendants of the independent variable are causally unrelated



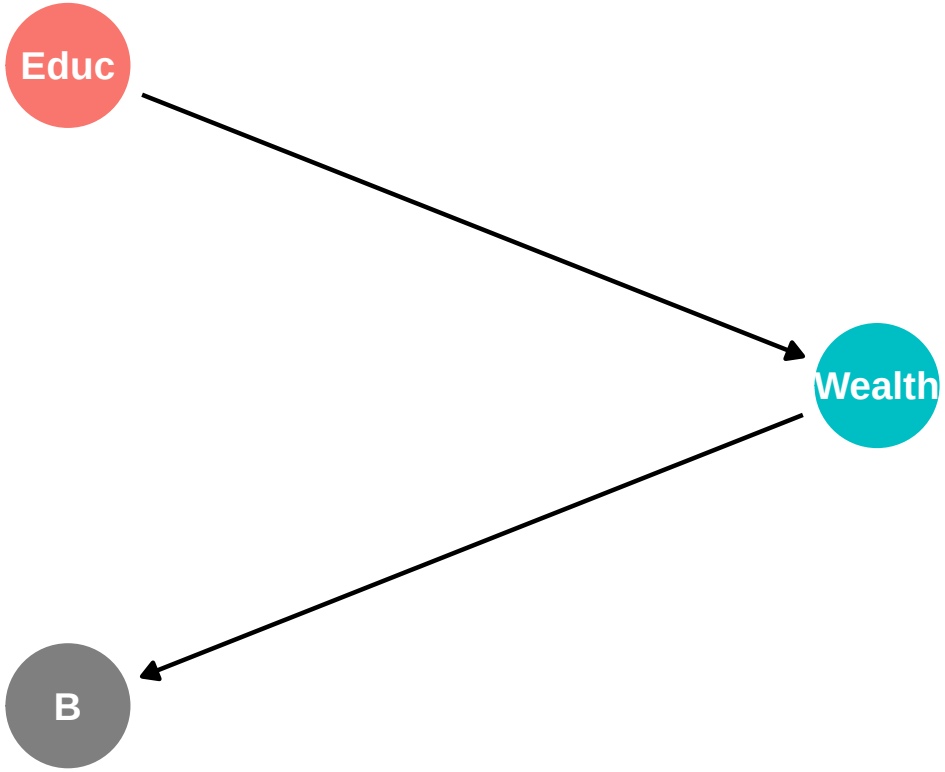
Good control

Independent causes of the outcome can increase precision



Bad control

Descendents of the outcome can induce bias



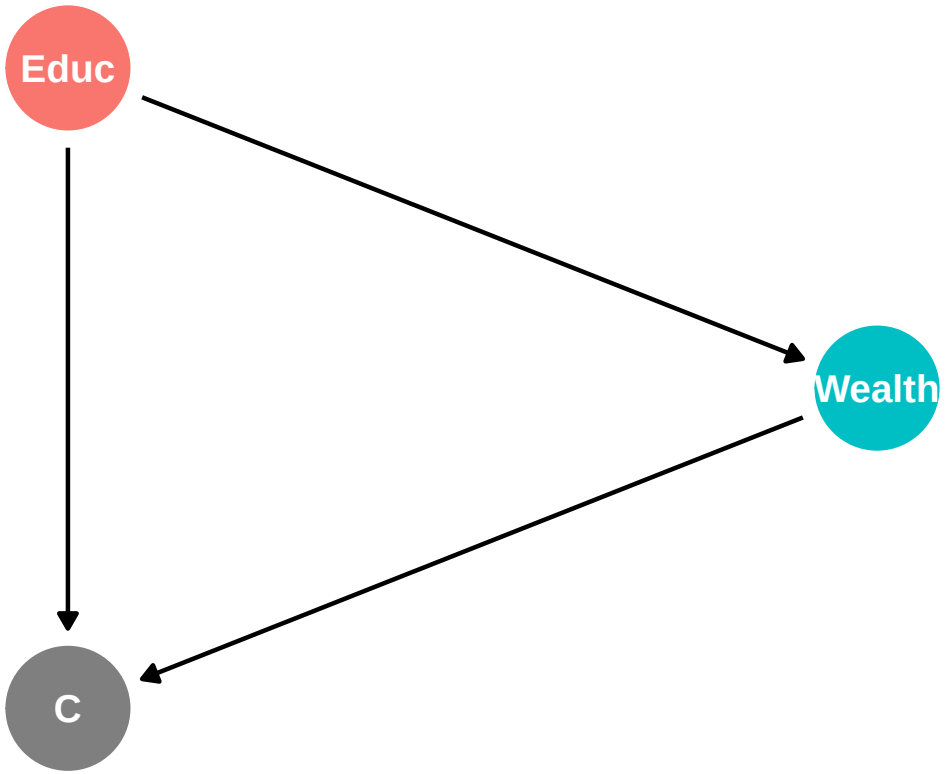
Variable selection

Colliders

- c is *caused by* y *and* x.
- In this case, c is a **collider** and controlling for c can introduce bias.

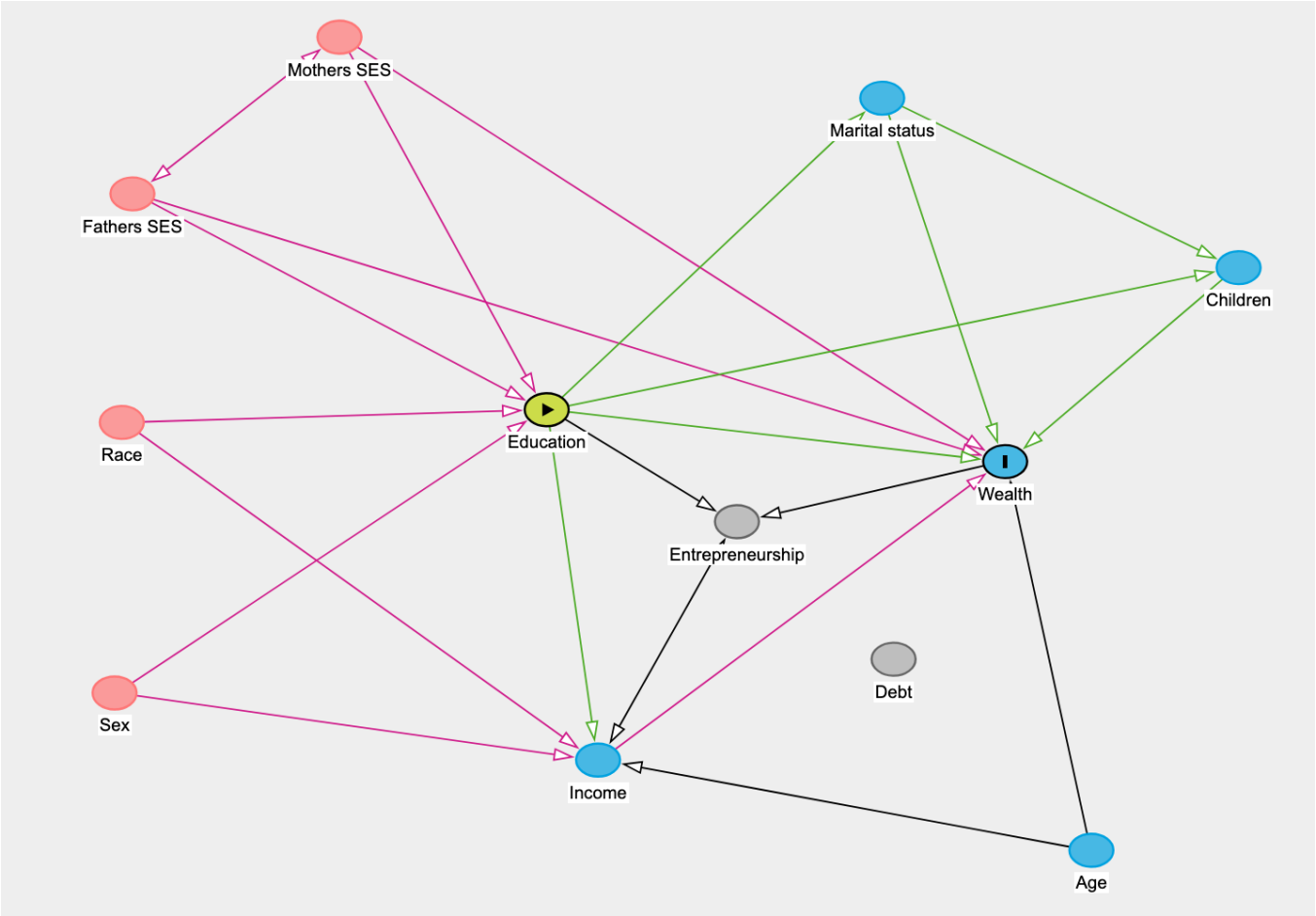
Colliders[7]

Don't condition on a collider



[7] See Elwert, Felix, and Christopher Winship. 2014. "Endogenous Selection Bias: The Problem of Conditioning on a Collider Variable." *Annual Review of Sociology* 40(1):31–53. doi: 10.1146/annurev-soc-071913-043455.

DAGs in the wild



Exercise

Drawing your DAG

1. Consider a relationship between two variables that you are interested in modeling
2. List some possible control variables
3. Based on knowledge and theoretical considerations, draw a DAG
   - Either by hand or using https://dagitty.net
4. What kinds of good and bad controls do you observe?

Multiple regression

Interpreting multiple controls

- So far all of the specifications have focused on a single predictor x and its relationship to the outcome y, net of controls
- It is common practice to give an interpretation to one or more, or even all, of the control variables
  - However, our focus is only on the identification of the main relationship, so these quantities may not be meaningful
- "Researchers should avoid providing any interpretation for estimates of control variables."

Keele et al. 2023, p.12.

Next week

Non-linear predictors

- Dummy variables
- Categorical variables
- Non-linear transformations

Lab

- Estimating and interpreting multiple regression models